

# ROCKETS

Rockets were invented in China a long time ago. They looked a bit like arrows and worked by burning gunpowder, which burns up very quickly, so the rockets did not travel very far. Since then, people have tried many ways of sending rockets up into space. Modern rockets usually use two liquid fuels. They mix together and burn.

Then the hot gas shoots out of the tail, pushing the rocket up and away.

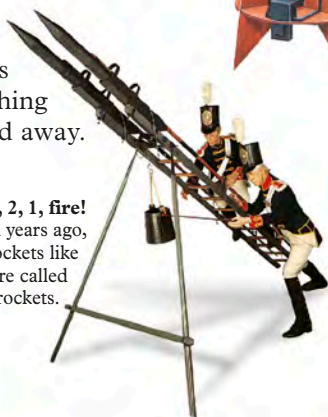


## See it go!

If you blow up a balloon and let it go without tying a knot in the neck, the air will rush out very quickly. When the air goes out one way, it pushes the balloon the other way—just like a rocket!

## 3, 2, 1, fire!

A hundred years ago, soldiers used rockets like this. They were called Congreve rockets.



## Saturn power

Saturn V is one of the biggest rockets ever built. It is as tall as a 30-story building! It was used in the Apollo program, which carried the first American astronauts to the moon.

*The stabilizing fins keep the rocket on course.*

*Five rocket engines*



V-2 Rocket  
1945



Gemini  
Titan 1964

## The Fly!

In 1931 Johannes Winkler launched his HW-1 rocket. It went 6.5 feet (two meters) into the air, turned over, and fell back to the ground. A month later he tried again and this time it climbed to 300 feet (90 meters) and landed 650 feet (200 meters) away.

## Up, up . . .

How far can you throw a ball? About 50 or 60 feet? It doesn't go on forever because the Earth's gravity pulls it back down again.